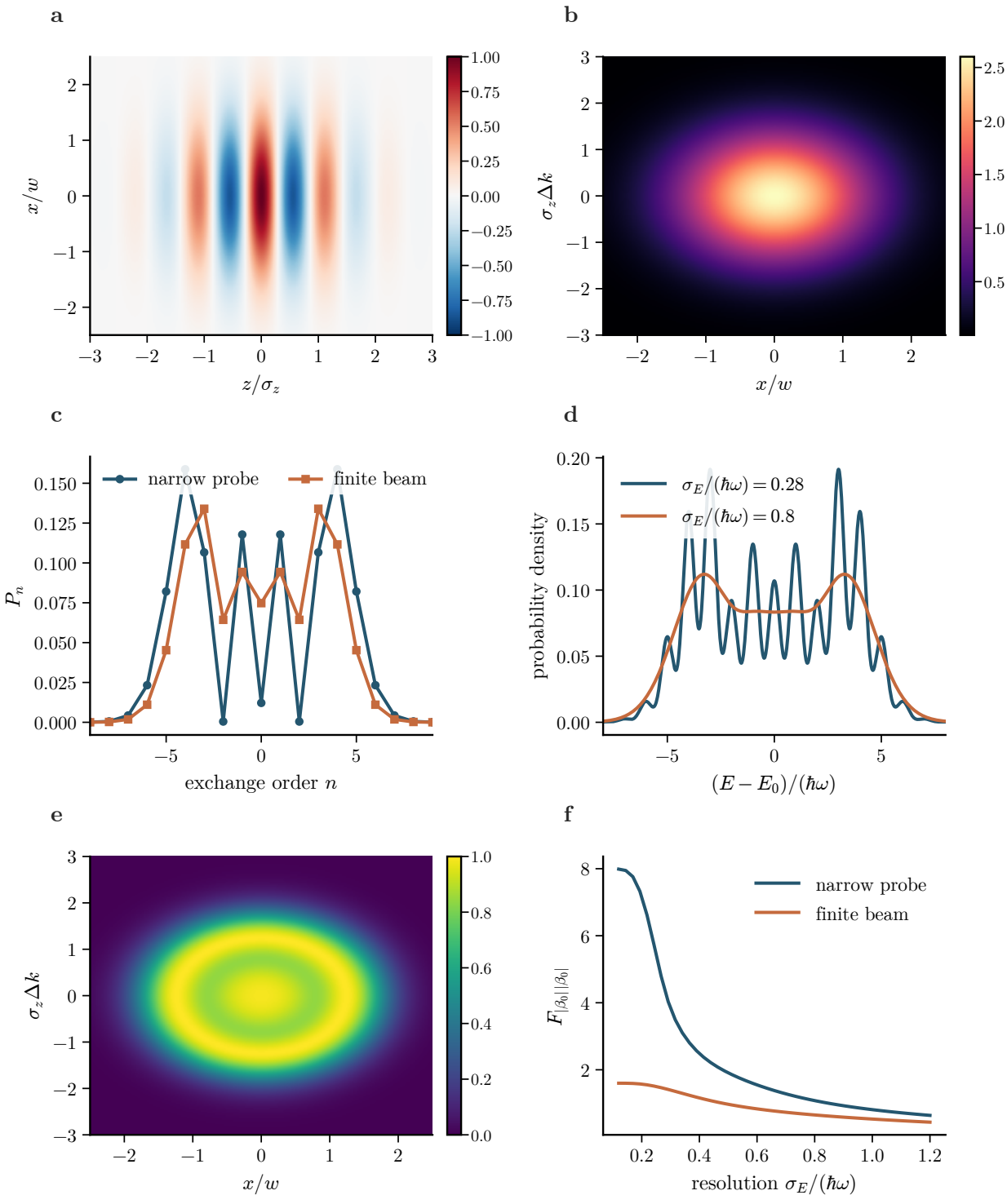

a
b
c
d
e
f
x/w
z/σ_z
$\sigma_z\Delta k$
P_n
exchange order n
narrow probe
finite beam
probability density
$\sigma_E/(\hbar\omega) = 0.28$
$\sigma_E/(\hbar\omega) = 0.8$
$(E - E_0)/(\hbar\omega)$
$F_{|\beta_0\rangle|\beta_0\rangle}$
resolution $\sigma_E/(\hbar\omega)$